

Phishing Website Classification

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DATS 2103 Data Mining

1. Introduction

The goal of this project is to build a binary classifier that predicts whether a given website is phishing or legitimate. We use the UCI Phishing Websites Dataset which contains 11,055 URLs with 30 integer-encoded features and a binary label. The features span three groups: address-bar properties such as URL length and presence of an IP address in the domain; HTML and JavaScript behaviors such as iframe usage and popup windows; and domain attributes such as SSL certificate state and Alexa traffic rank. Each feature is encoded as -1 for a phishing indicator, 0 for suspicious, or 1 for legitimate. We compare three models: Logistic Regression as a linear baseline from Chapter 4, and Random Forest and Gradient Boosting as ensemble methods from Chapter 8.

2. Data and Pre-processing

There are no missing values, all 31 columns are integer-typed, and the class split is near-balanced at 55.7% legitimate and 44.3% phishing. Of the 11,055 rows, 5,206 are exact feature-vector duplicates (47.1%), which we retain because they reflect the real distribution of URLs in the wild rather than data entry errors. Of the 5,785 unique feature patterns, 64 (~1.1%) appear with conflicting labels across rows, placing a soft accuracy ceiling of around 99% for any model trained on these features. A multicollinearity check revealed 26 feature pairs with $|r| > 0.5$, including Favicon and popUpWidnow at $r = 0.94$ and Shortining_Service and double_slash_redirecting at $r = 0.84$. This does not affect tree-based models but inflates standard errors in logistic regression.

For pre-processing, we recode the target from $\{-1, 1\}$ to $\{0, 1\}$ where 1 means phishing, following the scikit-learn convention. We apply an 80/20 stratified train/test split with a fixed random seed of 42, preserving the phishing rate in both subsets at approximately 44.3%. No normalization is needed since all features are already on the same $\{-1, 0, 1\}$ scale. The final dataset has $n = 11,055$ observations and $p = 30$ predictors, split into 8,844 training and 2,211 test observations.

3. Models and Results

We evaluate all three models using 5-fold stratified cross-validation on the training set, then refit on the full training set and assess on the held-out test set. Metrics reported are accuracy, AUROC, precision, recall, and F1, all in $[0, 1]$ where higher is better.

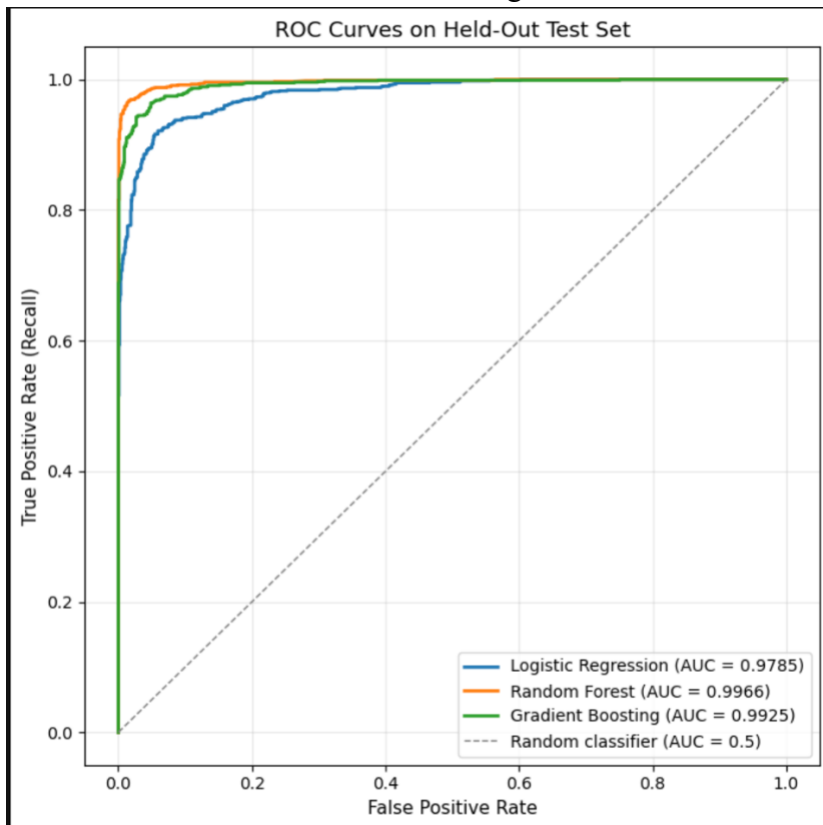
Random Forest is the best-performing model across every metric. In cross-validation it achieves a mean accuracy of 0.9695 (std = 0.0019) and a mean AUROC of 0.9945 (std = 0.0010). On the held-out test set it achieves 97.69% accuracy, an AUROC of 0.9966, precision of 0.9804, recall of 0.9673, and F1 of 0.9738. Gradient Boosting ranks second with a cross-validation accuracy of 0.9510 and AUROC of 0.9916, and a test accuracy of 95.66% with AUROC of 0.9925. Its AUROC nearly matches Random Forest but its accuracy trails by about 2 percentage points, suggesting a less optimal default threshold. Logistic Regression exceeds 92% accuracy in both settings but falls 4 to 5 points behind the ensembles, reflecting its inability to capture non-linear feature interactions without explicitly adding interaction terms.

Feature importance is consistent across all three models. SSLfinal_State is the most important feature in both tree models, contributing 32.2% of Random Forest importance and 71.5% of Gradient

Boosting importance. URL_of_Anchor ranks second at 24.6% for Random Forest and 14.5% for Gradient Boosting. These two features also have the highest Pearson correlations with the target at $r = 0.715$ and $r = 0.693$ respectively. A middle tier of web_traffic, having_Sub_Domain, and Links_in_tags appears consistently across all three models. In logistic regression, 21 of the 30 predictors are statistically significant at $\alpha = 0.05$, with URL_of_Anchor carrying the largest absolute coefficient at $|\beta| = 3.27$ and SSLfinal_State second at $|\beta| = 1.63$. The nine non-significant predictors correspond to features with near-zero univariate correlation such as Favicon, Iframe, and RightClick. Confusion matrices show roughly symmetric errors across all three classifiers, with no strong bias toward false positives or false negatives.

4. ROC Curve Analysis

The ROC curve plots the true positive rate against the false positive rate as the classification threshold varies. On the test set, Random Forest achieves an AUROC of 0.9966, Gradient Boosting 0.9925, and Logistic Regression 0.9785, all well above the random baseline of 0.5. The largest separation between models appears at low false positive rates, which is the most operationally important region since we want to minimize false alarms on legitimate sites.



5. Conclusions and Limitations

Phishing detection from URL and page-structure features is highly tractable. Random Forest approaches the practical accuracy ceiling at 97.69% test accuracy, and even the logistic regression baseline exceeds 92%. Tree ensembles outperform logistic regression by capturing non-linear interactions that the linear model cannot represent without explicit interaction terms. SSL certificate state and anchor-tag

destination dominate the signal across all three models and are also the features hardest for an attacker to forge at scale. We recommend Random Forest with 300 trees and default parameters.

A few limitations are worth noting. The dataset is from 2012 and may not reflect current phishing tactics. The 47% duplicate rate means random CV folds likely contain overlapping patterns in both training and test partitions, introducing a small optimistic bias. The 64 conflicting-label patterns set an irreducible error floor near 1%. And multicollinearity among several feature pairs reduces the reliability of individual logistic regression coefficient interpretations.

6. References

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